

Assignment - 3

→ Trends and Cloud Services Overview

1. Serverless Architecture Benefits:

Serverless Computing lets developers deploy functions without managing infrastructure. It scales automatically, reduces costs by charging per execution, and accelerates deployment cycles. Providers like AWS Lambda and Azure Functions are popular examples.

2. Progressive Web Apps (PWAs):

PWAs offer native app-like experiences through the browser. They are installable, work offline via Service and reach. They're ideal for mobile users who may not install apps from app stores.

3. Role of AI/ML in Architecture:

AI and ML are transforming architecture by enabling features like recommendation engines, fraud detection, and predictive analytics. They often integrate with Cloud Services like AWS SageMaker or Google Vertex AI for Scalable ML workflows.

→ Cloud Service Models:

• SaaS (Software as a service):

Delivers applications over the internet (e.g., Gmail, Zoom). Great for end-users needing ready-to-use solutions.

• PaaS (Platform as a service):

Provides a platform to develop, run and manage apps without infrastructure concerns (e.g., Heroku, Google App Engine)

- IaaS (Infrastructure as a service):

offers virtualized computing resources (e.g., AWS EC2, Azure VMs). Ideal for Sysadmins needing full control over VMs.